Introduction to Econometrics
Week 5

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Based on notes from previous instructors

March 2, 2026

Main learning outcomes

After studying this lecture, you should be able to:

- derive the GLS estimator and demonstrate that it is BLUE
- explain the differences between GLS and FGLS
- explain what an instrumental variable is
- explain the different sources of inconsistency and what IVs we can use to tackle them

Overview

1 Generalized least squares (GLS) 4

2 Introduction instrumental variables (IV) 13

2.1 What is an instrumental variable? 13

2.2 Omitted variable bias and presentations of the IV and 2SLS estimators 20

2.3 Measurement error in explanatory variables 29

2.4 Simultaneity 32

2.5 Some comments on IV-type estimators 37

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Generalized least squares (GLS)

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  1. we dropped the spherical error assumption allowing for heteroskedastic error terms

  2. we dropped the normality assumption

- these assumptions are not necessary for OLS estimator to be unbiased

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- but 1 implies:

  1.

  2.

Generalized least squares (GLS)

Under conditional heteroskedasticity, one strategy (the most popular nowadays) is therefore:

- to rely on large-sample theory for variance formulas and test statistics
- keeping in mind that OLS is still unbiased
- but not necessarily the estimator with the smallest variance in the class of unbiased linear estimators

Generalized least squares (GLS)

- an alternative is to use GLS
- for GLS we maintain the assumption used for small samples, including the normal error assumption
- but allow for conditionally heteroskedastic and correlated errors
- we can then transform the OLS model so that all assumptions in previous lectures apply to the transformed model
- the gain in doing so is that OLS estimation of the transformed model is once again BLUE
- therefore, there can be efficiency gains to using GLS rather than previous strategy

GLS

- we relax the spherical error assumption and assume that:

$$\mathbf{V}\left(\varepsilon|\mathbf{X}\right) = \sigma^2\mathbf{V}\left(\mathbf{X}\right)$$

  where $\mathbf{V}\left(\mathbf{X}\right)$ is non-singular and known (later, we relax this to make the following applicable). For simplicity of notation will denote it by $\mathbf{V}$

- notice that we allow not only for heteroskedasticity but also for serial correlation

- as $\mathbf{V}$ is non-singular, there exists a non-singular matrix $\mathbf{C}'$ s.t.

$$\underset{(n\times n)}{\mathbf{V}^{-1}} = \underset{(n\times n)}{\mathbf{C}'}\underset{(n\times n)}{\mathbf{C}}$$

GLS

1. transform the original linear model by multiplying by $\mathbf{C}$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \Rightarrow \underbrace{\mathbf{C}\mathbf{y}}_{\equiv\tilde{\mathbf{y}}} = \underbrace{\mathbf{C}\mathbf{X}}_{\equiv\tilde{\mathbf{X}}}\boldsymbol{\beta} + \underbrace{\mathbf{C}\boldsymbol{\varepsilon}}_{\equiv\tilde{\boldsymbol{\varepsilon}}} \Rightarrow \tilde{\mathbf{y}} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\boldsymbol{\varepsilon}}$$

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3. $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ obtained by estimating the transformed model by OLS is, therefore ? (i) and

Feasible GLS

- but we have assumed that we know $\mathbf{V}(\mathbf{X})$ in

$$\mathbf{V}\left(\varepsilon|\mathbf{X}\right) = \sigma\mathbf{V}\left(\mathbf{X}\right)$$

- but very unlikely that we do in practice
- so we will need to use an estimate $\hat{\mathbf{V}}$ of $\mathbf{V}$ instead
- the resulting estimator is called **feasible GLS**

One example of feasible GLS (known as WLS)

1. estimate the original model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ by OLS
2. compute the residuals: $u_i = y_i - \mathbf{x}_i'\boldsymbol{b}$
3. estimate $u_i^2 = \exp(\mathbf{z}\left(\mathbf{x}_i\right)'\boldsymbol{\gamma}) + \zeta_i$ where $\mathbf{z}$ is a function that selects only some of the elements of $\mathbf{x}_i$ (by non-linear least square for instance)
4. use $\hat{\mathbf{V}} = \text{diag}(\hat{u_i^2})$ where $\hat{u_i^2}$ are the predicted values of u_i^2 based on 3 (notice that here we assume heteroskedasticity but not correlation)

Some comments about FGLS

1. hard to get any results about FGLS in small sample, so nothing ensures that FGLS is BLUE for instance. More problematic, can be biased (and so not BLUE then).
2. for large samples, FGLS without correlated errors (called weighted least squares or WLS) is consistent even if the variance is misspecified
3. but in that case, nothing ensures that its asymptotic variance is smaller than OLS
4. efficiency gains require correct specification of heteroskedasticity (or at least close enough)
5. practice is to compute robust standard errors with FGLS as standard erros will be otherwise incorrect if the heteroskedasticity is mispecified

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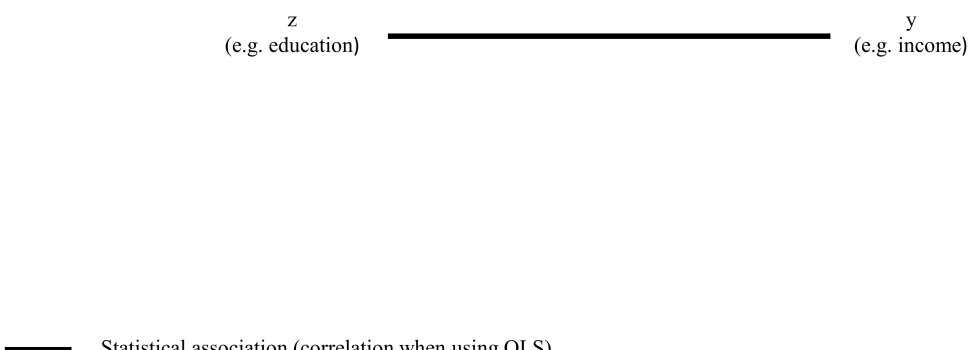
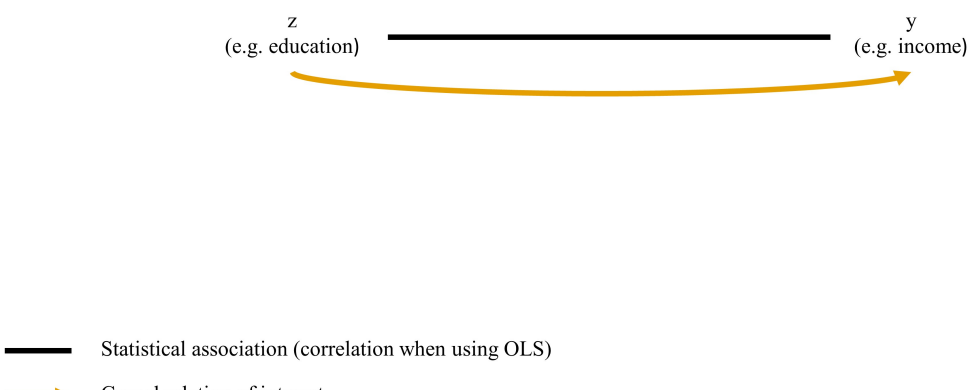
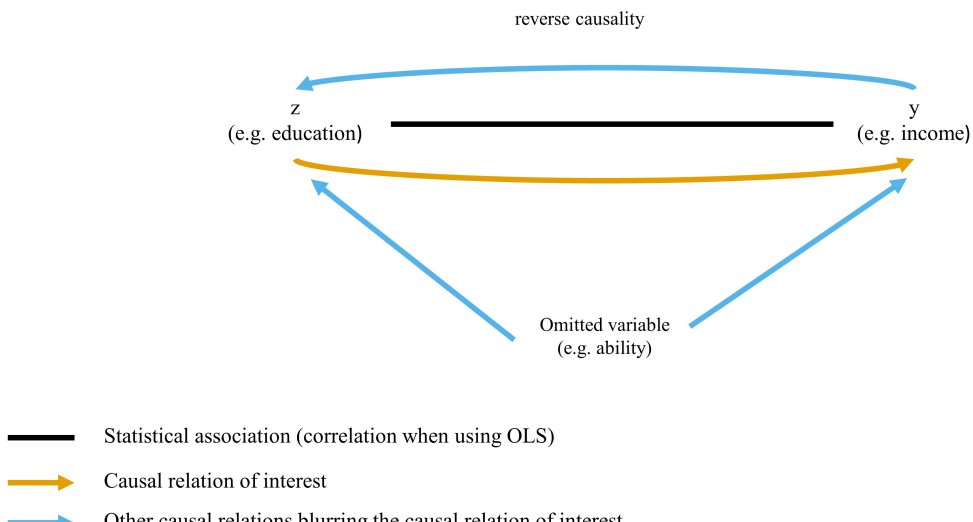


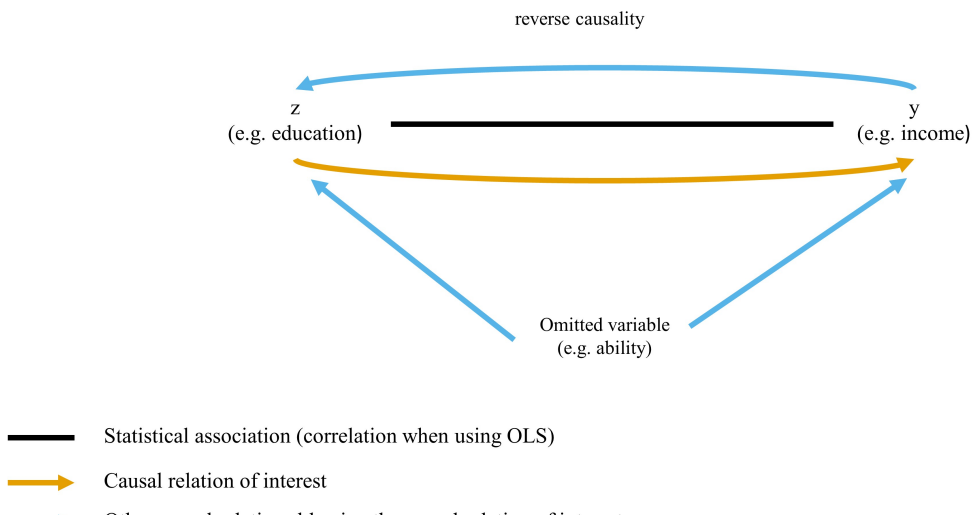
OLS (with intercept) is going to capture (asymptotically) all the statistical association (in a correlation sense) between $\mathbf{z}$ and y



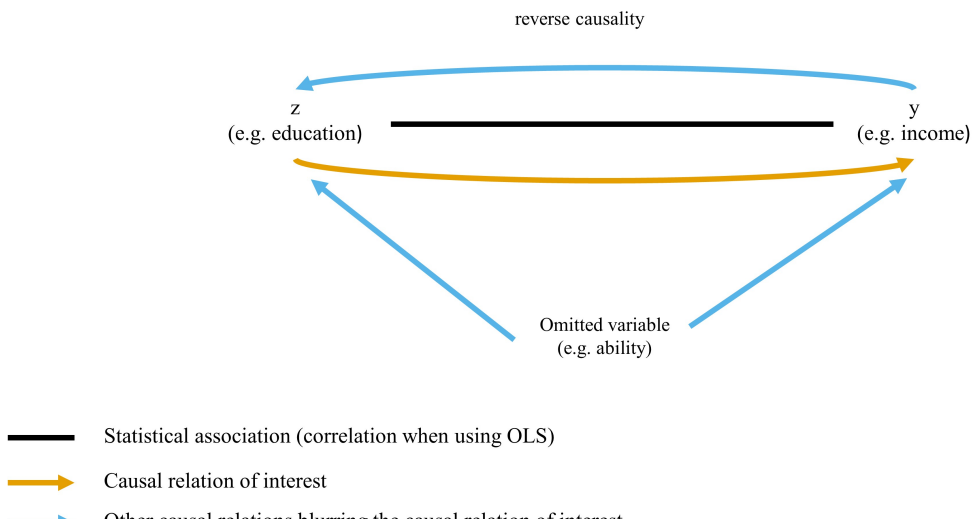
But we are not necessarily interested in all the statistical association between $\mathbf{z}$ and y. Our interest often lies in the causal effect of $\mathbf{z}$ on y.



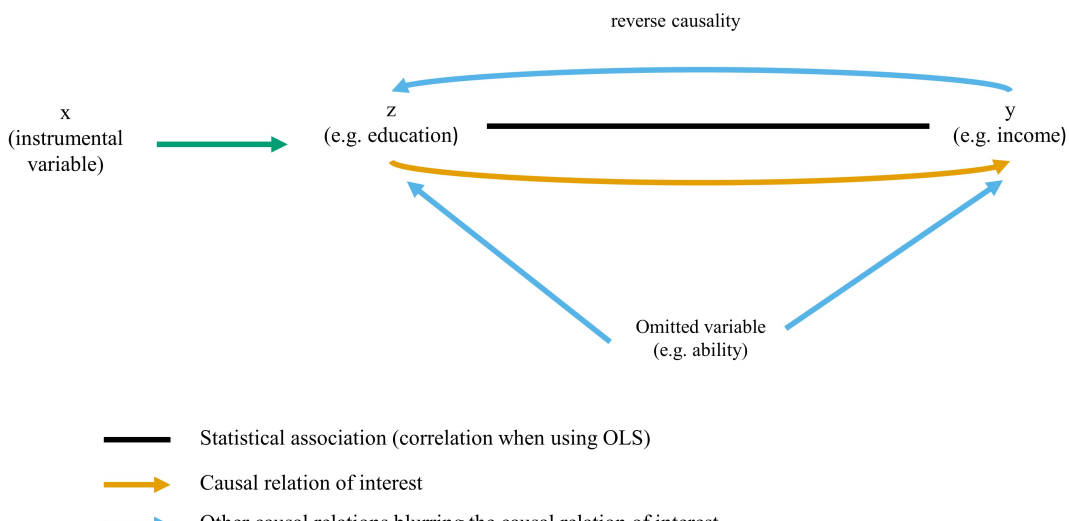
When looking at all the association between $\mathbf{z}$ and y, this causal effect can be blurred by other causal effects. Problem: OLS possibly captures all these.



Other way to understand this: observed correlation between $\mathbf{z}$ and y is the sum of all these effects. So, no way to discriminate between them by just looking at these correlations (unless we are sure that these other effects are absent). We need more information. What type?



Ideally if we could move $\mathbf{z}$ independently of the other variables and observe how y changes as a result, we would solve the problem.



This what an IV is! It is a variable that generates fluctuations in $\mathbf{z}$ independently of the other variables. Importantly IV must affect y only through its effect on $\mathbf{z}$.

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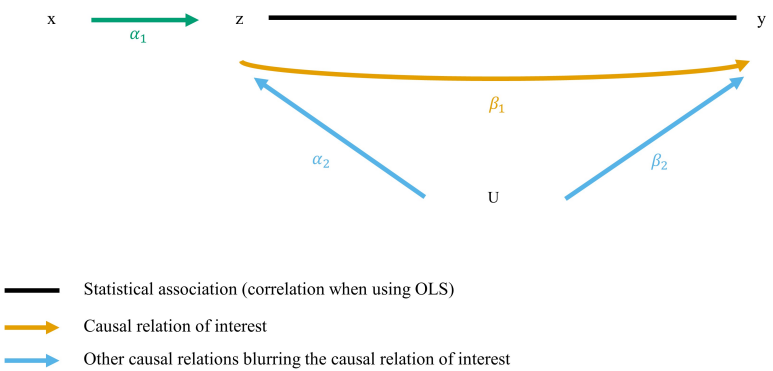
Omitted variable bias



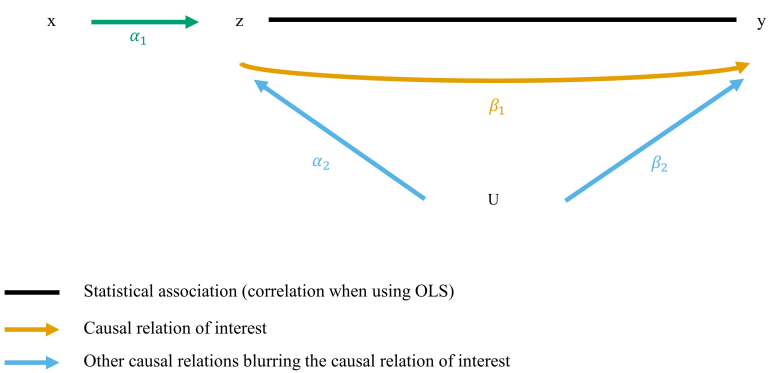
$$y = \beta_0 + \beta_1 z + \underbrace{\beta_2 U + \varepsilon}_{\equiv \tilde{\varepsilon}}$$

with

1. $cov(z, \varepsilon) = cov(U, \varepsilon) = 0$
2. $cov(z, U) \neq 0$

What is the implication if we try to estimate β_1 but we cannot control for U?

Omitted variable bias



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1. $cov(z, \varepsilon) = cov(U, \varepsilon) = 0$
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Suppose there is a variable x that satisfies:

1. $cov(x, U) = cov(x, \varepsilon) = 0$
2. $cov(x, z) \neq 0$

Compute $cov(y, x)$ using the expression of y above? If you observe x can you recover β_1 WITHOUT observing U?

Omitted variable bias

- suggests as a consistent estimator for β_1 (called IV estimator):

$$\hat{\beta}_{1,IV} = \frac{\text{sample covariance of } y \text{ and } x}{\text{sample covariance of } x \text{ and } z}$$

  if data are i.i.d. and sufficient assumptions are made for the necessary first moments to exist (remember LLN seen in lecture 4)

Omitted variable bias

$$cov\left(y,x\right) = \beta_1 cov\left(z,x\right) + \underbrace{cov\left(\beta_2 U + \varepsilon, x\right)}_{=0} \Rightarrow \beta_1 = \frac{cov\left(y,x\right)}{cov\left(x,z\right)}$$

- this example highlights the 2 key properties of an instrument:

  1. relevance: must be correlated with endogenous variable (after controlling for other regressors): here $cov\left(x,z\right) \neq 0$

  2. validity: must be uncorrelated with the error term in the equation for y, here: $cov\left(\beta_2 U + \varepsilon, x\right) = 0$; i.e. must affect y only through z

Omitted variable bias (2SLS)

- we could have computed another equivalent estimator, called two-stage least squares (2SLS), the idea is as follow.
- first stage: consider the LS projection of z on $(1, x)$. Denoting by (α_0, α_1) the LS projection coefficients, we have:

$$z = \alpha_0 + \alpha_1 x + \nu$$

with

$$cov(\nu, x) = 0$$

which implies $\alpha_1 =$ and $\alpha_0 =$.

Omitted variable bias (2SLS)

- Note by $\hat{z}$ the resulting predicted value of z based on x:

$$\hat{z} = \alpha_0 + \alpha_1 x$$

- second stage: consider the LS projection of y on $(1, \hat{z})$ and denote the LS projection coefficients (γ_0, γ_1), we then have:

$$y = \gamma_0 + \gamma_1 \hat{z} + \omega$$

with $cov(\hat{z}, \omega) = 0$ leading to

$$\gamma_1 =$$

Omitted variable bias (2SLS)

- for the model
$$y = \beta_0 + \beta_1 z + \tilde{\varepsilon}$$
with $cov(z, \tilde{\varepsilon}) \neq 0$ and the instrumental variable x such that $cov(x, z) \neq 0$ and $cov(x, \tilde{\varepsilon}) = 0$,

- we have shown that we can recover β_1 by projecting z on $(1, x)$, constructing the resulted predicted value $\hat{z}$, and projecting y on $(1, \hat{z})$.

- but we know that if our data are i.i.d., if the necessary expectations exist and if we can compute the OLS estimator (rank condition), OLS is always consistent for LS projection coefficients.

- so a consistent estimator for β_1 is obtained as what we just saw just by doing two OLS regressions instead of two projections:
  1. regress z on the x
  2. regress y on the predicted value of z based on x[1]

[1] $\hat{z}$ can be thought of as a subpart of z purged from its endogeneity

Omitted variable bias (2SLS)

- What are the assumptions needed to consistently estimate by OLS (α_0, α_1) in

$$z = \alpha_0 + \alpha_1 x + \nu$$

with $cov\,(\nu, x) = 0$ (by definition of LS projections coefficients)?

- What are the assumptions needed to consistently estimate by OLS (γ_0, γ_1) in

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with $cov(\hat{z}, \omega) = 0$ (by definition of LS projections coefficients)?

Remarks

- important: $\gamma_1 = \beta_1$ in the above derivations only if the instrument satisfies the properties of an instrumental variables
- notice that 2SLS will lead to a worse means-squared error than an OLS regression, why?
- once again the idea is that we need to find an instrumental variable (x) that generates variations in the endogenous variable (z)
- but generates movements in the dependent variable (y) only through its effect on the endogenous variable (z)
- if so, by keeping only the variations in the endogenous variable (z) that are caused by the IV (x), we get rid of endogeneity
- on top of endogeneity generated by omitted variable bias (as we just saw), IV can be used to tackle 2 other sources of endogeneity: measurement error and simultaneitaneity

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Source of endogeneity 2: measurement error in explanatory variables

- We want to estimate β in:
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$
where $E(\varepsilon_i x_i) = 0$.
- but we only observe $x_i^e = x_i + e_i$ with $E(e_i) = 0$, $cov(e_i, x_i) = 0$ and $cov(e_i, \varepsilon_i) = 0$
- what happens if we regress y_i on x_i^e instead of x_i?

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- what happens if we regress y_i on x_i^e instead of x_i?
- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \Rightarrow y_i = \beta_0 + \beta_1 x_i^e + (\varepsilon_i - \beta_1 e_i)$
- and $cov\left(x_i^e, \varepsilon_i - \beta_1 e_i\right) = cov\left(x_i + e_i, \varepsilon_i - \beta_1 e_i\right) = -\beta_1 var\left(e_i\right) \neq 0$
- so endogeneity issue

- OLS gives

$$\hat{\beta}_1 = \beta_1 \left(1 - \frac{var\left(e_i\right)}{var\left(x_i\right) + var\left(e_i\right)}\right) < \beta_1$$

- this is called attenuation bias
- intuition: the correlation between y and x is blurred by the measurement
- what is a good IV here?

- OLS gives

$$\hat{\beta}_1 = \beta_1 \left(1 - \frac{var\left(e_i\right)}{var\left(x_i\right) + var\left(e_i\right)}\right) < \beta_1$$

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- intuition: the correlation between y and x is blurred by the measurement
- what is a good IV here?
- a variable correlated with x_i^e who is uncorrelated with e_i, hence a variable that is correlated with x_i but uncorrelated with the measurement error e_i (e.g. log consumption could be instrumented by log wage)

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Source of endogeneity 3: simultaneity

- Consider demand and supply of coffee:

$$
\begin{aligned}
q_i^d &= \alpha_0 + \alpha_1 p_i + u_i \text{ (demand)} \\
q_i^s &= \beta_0 + \beta_1 p_i + \nu_i \text{ (supply)} \\
q_i^d &= q_i^s \text{ (market equilibrium)}
\end{aligned}
$$

  with $\alpha_1 < 0$ and $\beta_1 > 0$

- Assume $\mathbb{E}(u_i) = \mathbb{E}(\nu_i) = 0$ and cov $(u_i, \nu_i) = 0$

- and we seek to estimate the demand and supply curves separately

- Market equilibrium: $q_i = q_i^d = q_i^s$
- Solve for p_i and q_i:

$$p_i = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{\nu_i - u_i}{\alpha_1 - \beta_1}; q_i = \frac{\alpha_1\beta_0 - \alpha_0\beta_1}{\alpha_1 - \beta_1} + \frac{\alpha_1\nu_i - \beta_1 u_i}{\alpha_1 - \beta_1}$$

- Now p_i is endogenous in both equations:

$$\mathbb{E}\left(p_i u_i\right) = \text{cov}\left(p_i, u_i\right) = -\frac{\sigma_u^2}{\alpha_1 - \beta_1} > 0; \text{cov}\left(p_i, \nu_i\right) = \frac{\sigma_\nu^2}{\alpha_1 - \beta_1} < 0$$

- Regression of quantity on price estimates neither demand, nor supply curve

Source of endogeneity 3: simultaneity

- So-called simultaneity bias: regressor and error term are related through system of simultaneous equations
- This does not occur if $u_i = 0$ or $\nu_i = 0$

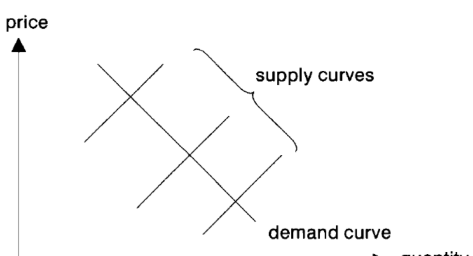

Figure. Identification of demand curve in absence of demand shifts

- Problem: we cannot disentangle demand and supply shifts
- Solution: find a variable that affects q_i *only* through either demand or supply
  - Demand-shifter identifies supply curve
  - Supply-shifter identifies demand curve
- Example supply-shifter: temperature in coffee-growing regions ($temp_i$)
- Likely to affect the endogenous coffee price (cov $(temp_i, p_i) \neq 0$)
- Unlikely to be related to the unobserved taste for coffee in NL (cov $(temp_i, u_i) = 0$)
- We can then identify demand curve parameters in the same way as we did for omitted variable (IV or 2SLS) – see ch3.1 Hayashi for details

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Cautionary notes on IV-type estimators

- IV-type estimators will generally be ill-behaved if instruments are weak even in quite large samples
- i.e. (assuming a constant) when instruments have low correlation with the endogenous variable(s)
- an intuitive way to understand this is to think of an instrument as sending a signal about the exogenous variations of the endogenous variable z
- if this signal (instrument) is weak then, an IV-type estimator will have difficulty picking up these exogenous variations and may well capture capture instead the other variations affecting the relation between y and z

Brief note on IV interpretation with heterogeneity

- so far, we have assumed homogenous effect of z on y, but what if effects are heterogeneous?
- OLS will capture the overall average effect of z on y (if OLS picks up a causal relationship)
- IV will capture the average effect of z on y for those for whom the instrument changes z ; often referred to as a local average treatment effect (LATE)
- if effect of z on y larger (in absolute value) for this group than for rest of population, IV-type estimator will generally produce larger estimates than OLS estimator
- question is whether this LATE is of interest for the application considered

Readings

Hayashi ch 1.6, 3.1 and 3.2